PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. Under the Panjab University Act VII of 1947 –enacted by the Govt. of India)

SYLLABI

Course Outline for

B.Voc. (Nutrition & Dietetics)

as per NEP 2020

w.e.f. Session 2024-25

Course Outline for B.Voc.(Nutrition & Dietetics) as per NEP 2020 w.e.f. session 2024-25

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
NAD 101	Core Discipline	Human Physiology	2	2	4	40	10	40	10	100
NAD 102	Core Discipline	Basics of Food and Nutrition	2	2	4	40	10	40	10	100
NAD 103	Core Discipline	Indigenous Food Systems	4	-	4	80	20	-	-	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 101	Multidisciplinary Course	Fundamentals of Information Technology[1]	3	-	3	60	15	-	-	75
SEC I	Skill Enhancement Course	SEC Options[2]			3					75
VAC I	Value-Added Course	VAC Options[3]			2					50
PW I	Project Work	Project Work-I & Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
NAD 201	Core Discipline	Food Microbiology	2	2	4	40	10	40	10	100
NAD 202	Core Discipline	Food Preservation and Storage	2	2	4	40	10	40	10	100
NAD_203	Core Discipline	Nutrition and Immunity	4	-	4	80	20	-	-	100
AEC / MIL	Ability Enhancement Course	English			2					50
B.Voc. MDSC 201	Multidisciplinary Course	Soft Skills and Personality Development[1]	3	-	3	60	15	-	-	75
SEC II	Skill Enhancement Course	SEC Options[2]			3					75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role –Food Microbiologist

* Summer Industrial Training of 2-4 weeks in a relevant Industry after 2nd Semester Examinations during summer break. The training report by the student is to be submitted within one week of the start of the 3rd Semester. Viva-Voce examination is to be held within three weeks of the start of the 3rd semester.

** Minimum Credits required to get Certificate Course in Nutrition & Dietetics = 48 + 4 #.

#Exit 1: Certificate Course-Student can exit after completion of 1st year having secured 48 Credits and will be awarded a

UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

1. Multidisciplinary Courses (MDSC): Annexure I
2. Skill Enhancement Courses (SEC): Annexure II
3. Value Added Courses (VAC): Annexure III

PROGRAMME OUTCOMES

PO1. Prepare qualified entry level professionals by providing expected domain knowledge.

PO2. Prepare a socially committed individual having high ethical values and empathy for the needs of society.

PO3. Design/development of solutions Develop innovative food products or substitutes or alternate solutions to create value and wealth for the betterment of the individual and society at large.

PO4. Problem analysis Identify, formulate, research literature, and solve nutritional deficiencies using fundamentals of clinical nutrition and dietetics, physiology, food science and biochemistry and relevant domain disciplines.

PO5. Professional Ethics Hold up and commit to professional ethics and ethical regulations, responsibilities, and norms of professional nutrition and dietetics practice.

PO6. Graduates are trained to develop innovative ideas and teams work skills to accomplish a common goal.

PO7. Produce lifelong learner with scientific skills for industrial applications and entrepreneurship.

Semester I

	COURSE TITLE	HUMAN PHYSIOLOGY
1.	**CourseCode**	NAD 101
2.	**Typologyof Course**	Core Discipline
3.	**Course Duration/ LP (Lecture) (Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100Marks (40THEORY + 10 IA) + (40 PRACTICAL+10IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
6.	**Job Role**	Exercise Physiologist
7.	**Course Objectives**	i) To provide a basic introduction to human physiology: the study of the functions of the human body and its constituent parts. ii) Apply knowledge of functional mechanisms and their regulation to explain the underlying physiology of common disease.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to: CO1. This course is based on basics of Anatomy and Physiology highlighting structural anatomy of various human body organs. CO2. To understand how these separate systems interact to yield integrated physiological responses. CO3. Students will be able to associate the structural anatomy with functions of body organs. CO4. The focus of this course is to evaluate the effect of alterations in structure on the functions of organs. CO5. The students under this course will able to understand the working of equipment's and instruments used in physiology lab. CO6. The course gives an insight about various parameters for diagnosis of illness.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT– I

Cell structure: Function and multiplication. Introduction to cell physiology and cell membrane
Tissues: Types, structure, and their functions.
Circulatory system: Anatomy and functions. Heart rate, cardiac cycle. Cardiac output. Blood composition and functions. Blood clotting, blood groups, radial pulse, blood pressure and factors affecting it.
Lymphatic system: Lymph glands and its function, spleen-structure, and functions.

UNIT– II

Skeletal System: Bones, joints (in brief).
Digestive system: Anatomy and functions of alimentary tract and accessory organs. Digestion and absorption of food.
Respiratory system: Anatomy, functions, mechanisms of external and internal respiration

UNIT– III

Endocrine System: List of endocrine glands, their hormones, functions, and their disorders (in brief)- Thyroid, parathyroid, adrenal, pancreas, pituitary.
Urinary system: Anatomy and functions. Formation and composition of urine.

UNIT-IV

Skin: Structure and functions
Reproductive system: Structure and functions of male and female reproductive organs. Menstruation and puberty.
Sensory organs: Eye, Ear, Nose, Tongue, skin – their function and basic structure (in brief).

Practical work:

1. Study of different types of microscopes and their uses.
2. Microscopic examination of permanent slides.
3. Identification of different blood groups.
4. Measurement of pulse and blood pressure.
5. Estimation of haemoglobin, RBC, WBC, TLC, DLC and ESR.

12. Suggested Resources for Reading

1. Keele, C.A. and Neil, E. Samson Wright ‘s Applied Physiology, Oxford University Press.
2. Tortora, G.J. and Anagnostakos, N.P. (1984). Principles of Anatomy and Physiology, Harper and Row Publisher.
3. Best, C.H. and Taylor, N.B., The Living Body, Asia Publishing House
4. Guyton, A.C., Functions of the Human Body, W.B. Saunders Co.
5. Wilson, K.J.W., Anatomy and Physiology in Health and Illness, ELBS/Churchill Livingstone.
6. Vidhya Ratan, Handbook of Human Physiology, New Central Book Agency

Note: Latest edition of text books may be used.

	COURSE TITLE	BASICS OF FOOD AND NUTRITION
1.	**CourseCode**	NAD 102
2.	**Typology of Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100Marks (40THEORY + 10 IA) + (40 PRACTICAL+10IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
6.	**Job Role**	Nutritionist
7.	**Course Objectives**	i) To elucidate the applied concepts of different areas of Foods and Nutrition. ii) To acquire knowledge about the nutritional needs and concerns of specific age groups/physiological conditions.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to CO1. The course focuses on the functions of food for healthy life. CO2. To designed to make the students understand different aspects of nutrients. CO3.To designed to make the students understand different aspects of nutrients. CO4. The students will be able to understand the nutrition information based on scientific reasoning for clinical, community, and food service applications. CO5. This practical course is designed make the students understand the importance of weights and measures in nutritional cookery. CO6. It is designed to know the nutritional value of basic recipes of each category through calculations.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT– I

Introduction to nutrition: Food as a source of nutrients, food pyramid, function of foods, definition of nutrition, nutrients, and energy; adequate, optimum and good nutrition, malnutrition.

Food guide – Basic five food groups and the nutritional contribution of each group in the diet.

Interrelationship between nutrition & health -Visible symptoms of good health

UNIT– II

Carbohydrates – Composition, functions, classification, food sources and requirements. Effect of deficiency and excess in the body. Role of fibre in human nutrition.

Fats & Oils - Composition, functions, classification, food sources and requirements. Effect of deficiency and excess in the body.

UNIT– III

Proteins- Composition, functions, classification, food sources and requirements. Effect of deficiency and excess in the body.

Water–role as a nutrient, functions, sources, requirement, water balance & effect of deficiency.

Vitamins – their functions, sources, requirements, effect of deficiency and excess of following vitamins:

- Fat Soluble - Vitamins A, D, E and K
- Water Soluble- Ascorbic acid and B-Complex: Thiamine, Riboflavin, Niacin, Pantothenic acid, Folic acid and B12.

UNIT-IV

Minerals –their functions and sources. Bioavailability, requirements, and effect of excess and deficiency of calcium, iron, iodine, sodium, potassium, and fluorine.

Functional foods.

Importance and methods of cooking and their effect on the appearance, nutritional value & taste**.**

Practical work:

1. Use and Care of different Kitchen equipment's
2. Weights and measures-Standard, Household measures for raw and cooked food.
3. Preparations from cereals (ANY TWO)- e.g. boiled rice, rice, pulao, chapati, puri, paratha, sandwiches, pastas, pancakes, cookies, and cake.
4. Preparations from pulses (ANY TWO)- e.g. dal makhani, sambar, chana masala, rajma etc.
5. Preparations from vegetables (dry, curries, koftas) (ANY TWO)- e.g. mixed vegetable, gobhi masala, vegetable kofta etc.
6. Preparations from milk (ANY TWO)- e.g. Porridge, curd/raita, paneer dishes, desserts and puddings custard, kheer, ice cream.
7. Soups (ANY TWO)- Basic, clear, and cream soups.
8. Salads with and without dressing. (ANY TWO)
9. Beverages (ANY TWO)- preparation of nourishing and refreshing beverages.
10. Snacks (ANY TWO) – pakoras, cutlets, cheese toast, upma, poha, peanut chikki, til-laddu preparations.
11. Use of leftover foods.

Requirements of all the nutrients should be according to RDA by NIN, ICMR

12. Suggested Resources for Reading

1. Guthrie, H., Andrews, Introductory Nutrition, St. Louis, Mirror /Mosby Collage.
2. Mudami S.R. &M. V Rajagopal, Fundamentals of Food and Nutrition, Wiley Eastern
3. Swaminathathan, M., Advanced Textbook of Foods, and Nutrition, Vol I, II, Bappco, Banglore.
4. Wilson, E.D., Principals of Nutrition, John Wiley &Sons
5. Swaminathathan, M, Principals of Nutrition and Dietetics, Bangalore Print & Publishing Co.
6. William, S., Nutrition and Diet Therapy, Mosby Co, St. Louis.
7. Bogert, L.J., & Calloway, D.H., Nutrition and Physical Fitness, W.B. Saunders & Co, Philadelphia.

Note: Latest edition of text books may be used**.**

	COURSE TITLE	INDIGENOUS FOOD SYSTEM
1.	**CourseCode**	NAD 103
2.	**Typologyof Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	60 hours of lectures for Theory (1 credit = 15 hours of theory learning based on 4 hrs./week) (1 Semester = 15 weeks)
4.	**Credits**	4 Credits
5.	**Max. Marks and Time**	100Marks (80THEORY+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
6.	**Job Role**	Ayurvedic physicians
7.	**Course Objectives**	i) To impart knowledge regarding promotion and conservation of medicinal plants ii) Apply knowledge regarding Ayurvedic medicinal plants and its impacts on health.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to CO1. This course describes both broad knowledge and in-depth relation of herbs and nutrition. CO2. It is based on study of different herbs and their use in day-to-day life. CO3. The course also relates nutrition and functional components of herbs by exploring the newer product development.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.
11.	**Course Content**	

UNIT– I

Ayurveda: Concept, history, classification and uses
Basic principles of Ayurveda, Allied systems of medicine in
Ayurveda. Concept of diet in Ayurveda.
Status of Ayurveda and Perspectives for its future application.

UNIT– II

Introduction to medicinal plants- Aloe Vera, Tulsi, Neem, Turmeric, Ginger, Moringa, Amalta and Brahmi; functional properties, traditional practices and nutritional significance Promoting sustainability, conservation, and utilization of medicinal plants.

UNIT– III

Role of herbs and spices in developed and developing countries.
Utility of medicinal plants in household remedies of curing various diseases
Functional Foods grown at household level and its role on human health; Factors influencing use of food for medicinal purposes.

UNIT-IV

Medicinal Plant market and industries in India.
Medicinal food products of the future - novel foods

12. Suggested Resources for Reading

1. Joshi, M.C., Text book of Indian Medicinal Plants –Scientific Publishers
2. Bhutya, R K., Handbook of Ayurvedic Medicinal Plants of India-Scientific Publishers
3. Kapoor, L.D., Handbook of Ayurvedic Medicinal Plants

Note: Latest edition of text books may be used.

	COURSE TITLE	PROJECT WORK
1.	**CourseCode**	PW I
2.	**Typologyof Course**	PROJECT WORK
3.	**Course Duration/ LP**	60 hours of practical (1 credit = 30 hours of practical based on 4 hrs./week)
4.	**Credits**	2 Credits
5.	**Max. Marks and Time**	50 Marks
6.	**Job Role**	Nutrition advisor
7.	**Course Objectives**	i) To create awareness about nutrition and healthy diet. ii) Identify and classify the food and foodstuff.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to CO1. Students will able to do research work. CO2.Students will be able to create awareness about how to guide or advise about healthy diet.
9.	**Pedagogy of the Course Work**	Practical
10.	**Instructions for paper setters and candidates**	Project submission and Viva
11.	**Course Content**	i) Survey to evaluate the dietary habits and assess the nutritional status of adolescent girls and middle-aged women, followed by an analysis and recommendation. ii) Proposing healthy and nutritious recipes tailored for these demographic group. iii) Submitting of project report.

Semester II

	COURSE TITLE	FOOD MICROBIOLOGY
1.	**CourseCode**	NAD 201
2.	**Typologyof Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100 Marks (80THEORY+10IA) + (40 PRACTICAL + 10 IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
6.	**Job Role**	Food Microbiologist
7.	**Course Objectives**	i) To study the origin of micro-organisms and their structure. ii) To understand the growth and activities of bacteria. iii) To gain knowledge of principles of various techniques used in prevention and control of micro- organisms in food. iv) To develop the knowledge about harmful bacteria, methods of transmission and anti-microbiology agents: antibiotics, germicides
8.	**Course Outcomes**	On successful completion of the course, the students will be able to CO1. Explain the interactions between microorganisms and the food environment, and factors influencing their growth and survival. CO2. Explain the significance and activities of microorganisms in food. CO3. Describe the characteristics of foodborne, waterborne and spoilage microorganisms, and methods for their isolation, detection, and identification. CO4. Explain why microbiological quality control programmes are necessary in food production.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT– I

Introduction of microbiology and its relevance to everyday life, general characteristics of bacteria, fungi, virus, protozoa and algae.

Growth of microorganisms: Growth curve, effect of environmental factors in growth of microorganism, pH, water activity, oxygen availability, temperature, and others.

UNIT– II

Microbiology of different foods: Spoilage and contamination, sources, types, effect on the following:

- Cereal and cereal products
- Sugar and sugar products
- Vegetables and fruits
- Meat and meat products
- Fish, egg, and poultry
- Milk and milk products
- Canned foods.

UNIT– III

Environmental microbiology:

- Water and water borne diseases
- Air and air borne diseases
- Soil and soil borne disease
- Sewage and diseases

Beneficial effect of microorganisms. Fermented foods: Bread, malt beverages.

UNIT– IV

Role of microorganisms in fermented milk products: Butter and cheese etc. Microbial enzymes in food processing. Relevance of microbial standards for food safety. Food hazard of microbial origin and occurrence and growth of microorganism in food.

Microbial intoxication and infections: Sources of contamination of food by pathogenic organisms, symptoms, and methods of control.

Practical work:

1. Study of equipment's in a microbiology lab.
2. Preparation of laboratory media and special media, cultivation and identification of bacteria, yeasts, and moulds.
3. Gram staining of bacteria.
4. Microbial examination of various foods: Foods and vegetables, meat, milk, and milk products.
5. Dye reduction test for assessing quality of milk.
6. Sterilization of various utensils.
7. Testing of water quality.
8. Development of aseptic conditions.
9. Visits to food processing units or any other organization dealing with advanced methods in food microbiology.

12. Suggested Resources for Reading

1. Frailer, W.C., Food Microbiology, McGraw Hill, New York
2. Kawata,K. Environmental Sanitation in India, Lucknow Publ. House
3. Pelezer, H.J. and Rober,D., Microbiology, McGraw Hill, New York
4. Banwart, G.T., Basic Food Microbiology, CBS Publ., New Delhi
5. Jay, J.H., Modern Microbiology, CBS Publ., New Delhi

Note: Latest edition of text books may be used.

	COURSE TITLE	FOOD PRESERVATION AND STORAGE
1.	**CourseCode**	NAD 202
2.	**Typologyof Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	30 hours of lectures for Theory + 60 hours of practical (1 credit = 15 hours of theory learning based on 2 hrs./week) (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 3-groups will be created for the purpose of Practical for each unit
4.	**Credits**	4 Credits [2+2]
5.	**Max. Marks and Time**	100Marks (40THEORY+10IA) + (40 PRACTICAL + 10 IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
6.	**Job Role**	Food technologists
7.	**Course Objectives**	i) To impart knowledge on the causes of food spoilage and methods of processing and preserving food. ii) To identify & select processing equipment and preservation methods appropriate for specific foods. iii) To describe the effects of methods of preservation on the quality of food.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to CO1. To understand the basic principles of food preservation CO2. To discuss the qualities and standards of available food preservatives and their suitability for different purposes. CO3. In this portion students learn about various techniques of preserving food products and detailed study of preserved food products, storage, nutritional aspects and food standards and food laws. CO4. students learn different methods of Food preservation – Drying, Freezing, Frying, canning, bottling etc. and preparation of jam, jelly, pickles, sauces, squashes.
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT– I

Definition, scope, and importance of food preservation.
Classification of food based on its shelf life.
Food spoilage: cause and effects.
Principles of Food preservation
Techniques of food preservation

UNIT– II

Methods of food preservation-Canning: Principles of Preservation of canned foods.
Thermal process for canned foods, spoilage in canned foods.
Water activity of food and its significance in food preservation;
Dehydration and drying of food items;
Low temperature preservation: cold storage, cold chain, freezing (including cryogenic freezing)
Preservation by fermentation: curing and pickling;
Non-thermal (e.g. high-pressure processing)
Ionization radiation; Use of preservative in foods:
Chemical preservative, bio preservatives, antibiotics, lactic acid bacteria

UNIT– III

Selection and purchase of foods for preservation.
Storage of common perishable, semi-perishable and non-perishable foods.
Storage stability of selected Frozen Foods.

UNIT-IV

Packaging and packing materials, labelling, and costing of products.
Food additives, preservatives, anti-oxidants, colouring matter, and emulsifying agents.

Practical work:

1. Preparation and storage of:
   i) Squashes and syrups
   ii) Jams and jellies
   iii) Marmalades and other fruit preserves
   iv) Ketchup/sauces and chutneys
   v) Pickles (with and without oil).
2. Preparation of items utilizing cereals and legumes and their storage.
3. Demonstration of canning and bottling of fruits and vegetables.
4. Demonstration of Sealing Process used in Food Industry
5. Visit to food industry.

12. Suggested Resources for Reading

1. Desrosier, L.P, The Technology of Food Preservation, CBS Publishers & Distributors
2. Potter, N.N & Hotchkiss, J.H., Food Science, CBS Publishers & Distributors
3. Srivastava, R.P. & Kumar, S., Fruits and Vegetable Preservation, Principle and Practices, CBS Publishers & Distributors
4. Muller, H.G., Fundamentals of Foods and Nutrition. Wiley Eastern Pvt Ltd.
5. Modern Technology on Food preservation by NIIR Board.

Note: Latest edition of text books may be used.

	COURSE TITLE	NUTRITION AND IMMUNITY
1.	**CourseCode**	NAD 203
2.	**Typologyof Course**	Core Discipline
3.	**Course Duration/ LP (Lectures) (Practical)**	60 hours of lectures for Theory (1 credit = 15 hours of theory learning based on 4 hrs./week) (1 Semester = 15 weeks)
4.	**Credits**	4 Credits
5.	**Max. Marks and Time**	100Marks (80THEORY+20IA) i) The exam of this course will be of 3hours duration. ii) There will be no objective type questions. iii) Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. iv) Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.
6.	**Job Role**	Nutrition consultant
7.	**Course Objectives**	Examining how nutrition reduces the side effects of medical treatments in general.
8.	**Course Outcomes**	On successful completion of the course, the students will be able to CO1. To provide knowledge about various foods and their quality and how to use them to provide immunity to the human body. CO2. The immune system can boost some components, such as vitamins, minerals, antioxidants, and phenolic compounds, because these components aid in forming T cells, B cells, and antibodies and enhance the function of melanoproteins and neutrophils. CO3. To understand the role of functional food ingredients of various foods and their mechanism. CO4. To illustrate the health benefits of using nutraceuticals
9.	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
10.	**Instructions for paper setters and candidates**	i) The syllabus of this paper has been divided into FOUR units. ii) Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. iii) The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. iv) All questions carry equal marks.

11. Course Content

UNIT– I

Definition of immunity, Immune system: The defence mechanisms of the body.
Role of nutrients in immune functions. Under nutrition-causes and consequences.

UNIT– II

Infection: Risk of infection, influence of infections on nutrition and susceptibility to infection.
Role of nutrients in prevention of risk of infections.
Childhood mortality associated with infectious diseases of gastro-intestinal tract and lungs.

UNIT– III

Age related immune depression.
Role of nutraceuticals and functional foods.
Nutrition in HIV/AIDS and tuberculosis.

UNIT-IV

Probiotics, prebiotics, and immunity
Food allergy (with emphasis on wheat allergy, peanut, lactose intolerance)
Latest health concerns

12. Suggested Resources for Reading

1. Barasi, M. E., Human Nutrition-A health perspective. Arnold International Student ‘s edition.
2. Gershwin, M.E., Nestel, P. and Keen, C.L., Handbook of Nutrition and Immunity. Humana Publishers
3. Owen, J., Punt, J. and Stanford, S., Kuby Immunology: International Edition. Macmillan Publishers.
4. Prason, S.K. and Poddar, T., Perfect Health-Body Diet and Nutrition. V & S Publishers.

Note: Latest edition of text books may be used.